

Neurodevelopmental Disorder with or without Anomalies of the Brain, Eye, or Heart (NEDBEH): A Comprehensive Disease Characteristics Report

Disease: Neurodevelopmental Disorder with or without Anomalies of the Brain, Eye, or Heart (NEDBEH) **OMIM:** #616975 · **MONDO:** MONDO:0014744 · **Gene:** *RERE* (1p36.23) · **Category:** Mendelian, autosomal dominant

Evidence base: aggregated human clinical case series plus zebrafish, mouse, and in-vitro functional studies. All content is literature-derived (no primary patient dataset was provided).

Summary

Neurodevelopmental Disorder with or without Anomalies of the Brain, Eye, or Heart (NEDBEH), also called *RERE*-related disorder, is an ultra-rare autosomal-dominant Mendelian condition caused by heterozygous pathogenic variants in *RERE* (arginine-glutamic acid dipeptide repeats gene), which encodes the nuclear receptor coregulator Atrophin-2. Most disease-causing variants arise *de novo*. The gene lies within the proximal 1p36 critical region, and the NEDBEH phenotype substantially overlaps that of the far more common 1p36 deletion syndrome (~1 in 5,000 newborns). Affected individuals near-universally show developmental delay, intellectual disability, and/or autism spectrum disorder, accompanied by a variable spectrum of structural anomalies of the brain, eye, heart, kidney, ear, and palate ([PMID: 27087320](#)).

Mechanistically, *RERE*/Atrophin-2 is a widely expressed nuclear receptor coregulator that positively regulates retinoic acid (RA) signaling through the chromatin-associated WHHERE complex (WDR5, HDAC1, HDAC2, *RERE*) together with the histone methyltransferase EHMT2/G9a, and it also modulates Sonic hedgehog (Shh) signaling. NEDBEH-associated variants behave as hypomorphs in repressing shh signaling, and a genotype–phenotype axis has emerged: loss-of-function (haploinsufficiency) variants cause milder, sometimes-inherited

disease with fewer multisystem anomalies, whereas missense variants and in-frame duplications affecting the histidine-rich region (HRR) of the Atrophin-1 domain produce more severe multisystem phenotypes—including a CHARGE-syndrome-like presentation—and appear to act by a gain-of-function or dominant-negative mechanism (PMID: 29330883; PMID: 41988794).

Diagnosis is molecular—via whole-exome/whole-genome sequencing, multigene neurodevelopmental panels, or chromosomal microarray—with no biochemical biomarker or newborn screening test. Fewer than ~60 individuals have been reported worldwide. There is no disease-specific or curative therapy; management is supportive, multidisciplinary, and increasingly variant-specific, guided by genetic counseling. Animal models (*RERE*-deficient zebrafish and mice) faithfully recapitulate key features, and *Shh* pathway inhibition (*HPI-1*) rescues coloboma in zebrafish—identifying a preclinical therapeutic lead (PMID: 36576487).

1. Disease Information

Overview. NEDBEH is a Mendelian neurodevelopmental disorder defined by a near-constant core of developmental delay (DD), intellectual disability (ID), and/or autism spectrum disorder (ASD), plus a variable constellation of congenital anomalies affecting the brain, eye, and heart (as well as kidney, ear, and palate). It was first delineated by Fregeau et al. (2016), who described ten individuals carrying rare, putatively damaging *RERE* variants and noted that the associated defect spectrum overlaps that of proximal 1p36 deletions: *"In this report, we describe ten individuals with intellectual disability, developmental delay, and/or autism spectrum disorder who carry rare and putatively damaging changes in RERE"* (PMID: 27087320).

Key identifiers: - **OMIM:** #616975 (Neurodevelopmental disorder with or without anomalies of the brain, eye, or heart) - **MONDO:** MONDO:0014744 - **Gene:** *RERE*, HGNC:9965, chromosome 1p36.23 - **ICD-11:** neurodevelopmental disorder categories (no NEDBEH-specific code); related monosomy 1p36 is separately coded - **MeSH:** No dedicated descriptor; indexed under intellectual disability / developmental disabilities

Synonyms / alternative names: *RERE*-related disorder(s); *RERE* syndrome; NEDBEH.

Information source. Knowledge is derived predominantly from aggregated disease-level resources (OMIM, published cohort series) and individual case reports, not from large EHR datasets. The largest assembled series to date comprises 54 individuals (PMID: 41988794).

2. Etiology

Primary cause — genetic. NEDBEH is caused by heterozygous pathogenic variants in *RERE*. In the founding cohort, *"In all cases in which both parental DNA samples were available, these changes were found to be de novo"* (PMID: 27087320). *RERE* lies in the proximal 1p36 critical region; large 1p36 deletions encompassing *RERE* cause an overlapping multisystem phenotype.

Genetic risk factors. The disease is monogenic: the causal factor is a pathogenic *RERE* variant. Variant class is the principal modifier of risk for multisystem involvement—HRR missense and in-frame HRR duplications confer higher risk of structural eye defects, congenital heart defects, renal anomalies, and sensorineural hearing loss than loss-of-function variants (PMID: 29330883).

Environmental risk / protective factors. No environmental, lifestyle, occupational, or infectious risk or protective factors have been established for this monogenic disorder. Because *RERE* is a coregulator of retinoic acid signaling, retinoic-acid biology is mechanistically central, but no dietary or exposure-based modifier has been validated in patients.

Gene–environment interactions. None established. The disorder is defined by its genetic etiology; the "environment" of relevance is the developmental/signaling milieu (RA and Shh gradients) rather than external exposures.

3. Phenotypes

The core phenotype—developmental delay, intellectual disability, and/or autism spectrum disorder—is present across essentially all variant groups: *"Developmental delay, intellectual disability, and/or autism spectrum disorder were prevalent across all groups"* (PMID: 41988794). Recurrent associated features reported by Fregeau et al. included *"hypotonia, seizures, behavioral problems, structural CNS anomalies, ophthalmologic anomalies, congenital heart defects, and genitourinary abnormalities"* (PMID: 27087320).

Phenotype	Type	Approx. frequency	Onset	Suggested HPO term
Developmental delay	Neurodevelopmental	Near-universal	Infancy/congenital	HP:0001263
Intellectual disability	Neurodevelopmental	Very frequent	Childhood	HP:0001249
Autism spectrum disorder	Behavioral	Frequent	Childhood	HP:0000717
Hypotonia	Clinical sign	Frequent	Neonatal/infancy	HP:0001252
Seizures	Clinical sign	Common	Childhood	HP:0001250
Structural CNS anomalies	Physical	Common	Congenital	HP:0002011
Ophthalmologic anomalies (coloboma, optic atrophy, refractive error)	Physical	~1/3	Congenital	HP:0000478 / HP:0000589 / HP:0000648
Congenital heart defect (e.g., VSD)	Physical	Common (enriched in HRR missense)	Congenital	HP:0001627 / HP:0001629
Renal/genitourinary anomalies	Physical	Variable	Congenital	HP:0000119
Sensorineural hearing loss	Clinical sign	Variable (enriched in HRR missense)	Congenital/childhood	HP:0000407
Cleft palate/lip (orofacial clefting)	Physical	Variable	Congenital	HP:0000175
Chiari type I malformation	Physical	Rare (single case)	Congenital	HP:0002308

Phenotype	Type	Approx. frequency	Onset	Suggested HPO term
Dysmorphic facial features	Physical	Variable	Congenital	HP:0001999

Phenotype characteristics. Onset is congenital/early-childhood. Severity is variable and stratified by variant class: HRR missense/in-frame duplications produce more severe, multisystem disease, while loss-of-function variants are milder and can even be inherited from a mildly-symptomatic or asymptomatic parent (PMID: 41988794). The neurodevelopmental phenotype is generally **non-progressive/stable**; a Chiari-I case reported progressive spasticity but is exceptional (PMID: 41669386).

Quality of life. Not formally quantified with EQ-5D/SF-36 in published cohorts. Impact derives from lifelong ID/DD (education, independent living), ASD-related behavioral needs, seizure burden, and the surgical/medical needs of congenital anomalies.

4. Genetic / Molecular Information

Causal gene. *RERE* (arginine-glutamic acid dipeptide repeats; Atrophin-2), HGNC:9965, chromosome **1p36.23**. *RERE* is "a widely-expressed nuclear receptor coregulator that positively regulates retinoic acid signaling" (PMID: 27087320).

Pathogenic variants. - **Variant types:** missense (notably within the Atrophin-1 domain / histidine-rich region), in-frame duplications (a recurrent two-amino-acid duplication), and loss-of-function alleles (nonsense, frameshift, splice, whole-gene deletion via 1p36 loss). - **Classification:** pathogenic / likely pathogenic per ACMG/AMP; the 54-individual cohort also included variants of uncertain significance (PMID: 41988794). - **Origin:** germline; usually **de novo**, but inherited variants exist—the first inherited variant was reported in a patient and his mildly affected mother (PMID: 36053530). - **Functional consequence:** Two mechanisms. Loss-of-function → **haploinsufficiency**. HRR missense variants are structurally stabilizing: "*HRR missense variants were structurally stabilizing, suggesting a gain-of-function or dominant-negative mechanism*" (PMID: 41988794).

Genotype–phenotype correlation (central molecular finding). "*Point mutations affecting the Atrophin-1 domain of RERE are associated with an increased risk of structural eye defects, congenital heart defects, renal anomalies, and sensorineural hearing loss when compared with*

loss-of-function variants that are likely to lead to haploinsufficiency", and "We have also identified a recurrent two-amino-acid duplication in this region that is associated with the development of a CHARGE syndrome-like phenotype" (PMID: 29330883). The 2026 cohort confirmed that "Loss-of-function variants are associated with fewer multisystem anomalies than missense variants and are more likely to be inherited from a mildly symptomatic or asymptomatic parent" (PMID: 41988794).

Convergent HX/HRR motif biology. The histidine-rich (HX-repeat) motif is shared across non-homologous genes: disruption of the analogous HX-repeat motif in *ATN1* (Atrophin-1) causes CHEDDA (congenital hypotonia, epilepsy, developmental delay, digit abnormalities), a recognizable non-progressive neurocognitive syndrome distinct from polyglutamine DRPLA. The authors note *"non-homologous human genes containing similar motifs... including RERE, where disruptive variants in this motif have also been linked to a separate condition"*—reinforcing that perturbation of the HX motif itself is pathogenic (PMID: 30827498).

Modifier genes. In 1p36 deletion syndrome, co-deleted genes modify severity (e.g., *KCNAB2*, *CHD5* for neurodevelopmental prognosis; *PRDM16*, *PRKCZ*, and *RERE* implicated in cardiac complications) (PMID: 25172301). For isolated *RERE* variants, variant class (LoF vs HRR missense) is the dominant modifier.

Epigenetic dimension. RERE functions within chromatin-modifying complexes (with HDAC1/HDAC2 and the H3K9 methyltransferase EHMT2/G9a), placing NEDBEH among disorders of epigenetic regulation of development (PMID: 28959017; PMID: 29300383).

Chromosomal abnormalities. Proximal 1p36 deletions that encompass *RERE* produce an overlapping multisystem phenotype and are detected by chromosomal microarray/FISH (PMID: 27087320; PMID: 25172301).

5. Environmental Information

No environmental, lifestyle, or infectious contributors are established for NEDBEH. It is a monogenic developmental disorder. The mechanistically relevant "chemical" context is endogenous **retinoic acid** (CHEBI:26536) signaling, which RERE coregulates, and **Sonic hedgehog** pathway output—not exogenous exposures. No toxin, radiation, occupational, dietary, or pathogen association has been reported.

6. Mechanism / Pathophysiology

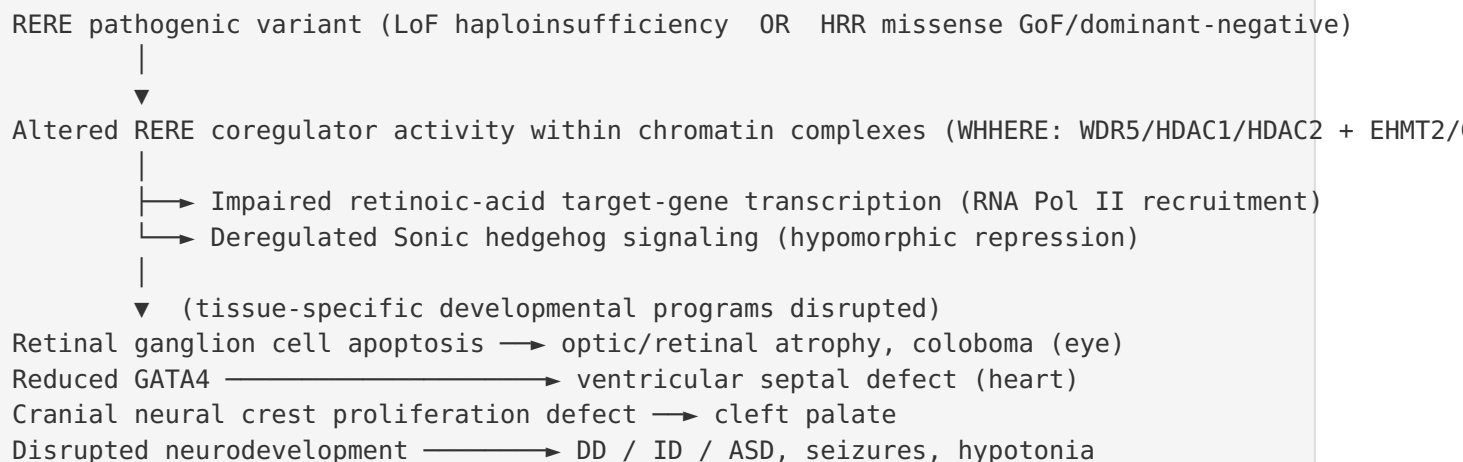
Molecular pathways

RERE/Atrophin-2 is a nuclear receptor coregulator of **retinoic acid (RA) signaling**. Vilhais-Neto et al. identified the **WHHERE complex** as required for RA-dependent regulation of embryonic bilateral symmetry: *"we identify a protein complex, containing Wdr5, Hdac1, Hdac2 and Rere (named WHHERE), which regulates RA signaling and controls embryonic symmetry"*, and *"Upon RA treatment, WHHERE and Ehmt2 become enriched at RA target genes to promote RNA polymerase II recruitment"* (PMID: 28959017). RERE additionally regulates **Sonic hedgehog (Shh)** signaling; NEDBEH-associated variants *"function as hypomorphs in their ability to repress shh signaling and some exhibit abnormal nuclear localization"* (PMID: 36576487).

Organ-specific causal chains

- **Eye (coloboma / optic atrophy).** *"RERE deficiency causes progressive loss of retinal cells and apoptosis of retinal cells in the ganglion cell layer as early as E17.5"*, producing retinal and optic-nerve atrophy (PMID: 33742727). In zebrafish, coloboma arises from optic-fissure closure defects driven by deregulated shh signaling, and *"Inhibiting shh signaling by the protein inhibitor HPI-1 rescues coloboma"* (PMID: 36576487).
- **Heart (ventricular septal defect).** *"RERE deficiency leads to decreased expression of GATA4 and the development of ventricular septal defects"* (PMID: 30061196).
- **Palate (orofacial clefting).** *"ablation of Rere in cranial neural crest (CNC) cells, mediated by a Wnt1-Cre, leads to delayed elevation of the palatal shelves and cleft palate"* (PMID: 33772547).
- **Brain/behavior.** Broadly expressed RERE and its role in RA-dependent neurodevelopment underlie the near-universal DD/ID/ASD; specific structural CNS anomalies and (rarely) Chiari I are reported (PMID: 27087320; PMID: 41669386).

Upstream → downstream logic



Cellular processes, GO / CL / cell-type terms

- **Biological processes (GO):** retinoic acid receptor signaling pathway (GO:0048384); smoothened/Shh signaling (GO:0007224); regulation of transcription by RNA polymerase II (GO:0006357); histone modification (GO:0016570); apoptotic process (GO:0006915); neural crest cell development (GO:0014032); heart development (GO:0007507); camera-type eye development (GO:0043010).
- **Cellular component (GO):** nucleus (GO:0005634); nuclear body (GO:0016604) — consistent with abnormal nuclear localization of some variants.
- **Cell types (CL):** retinal ganglion cell (CL:0000740); cranial neural crest cell (CL:0000333); cardiac myocyte / cardiac neural crest derivatives (CL:0000746).

Metabolic, immune, biochemical

No specific metabolic derangement, enzyme deficiency, or immune/autoimmune involvement is described. The biochemical defect is dysregulated **transcriptional coregulation** (RA/Shh), not a metabolic block.

Molecular profiling

No large-scale patient transcriptomic/proteomic/metabolomic datasets are published for NEDBEH. Mechanistic evidence is primarily from mouse and zebrafish models and cell-based assays (e.g., GATA4 downregulation; shh-reporter hypomorphism).

7. Anatomical Structures Affected

- **Organ level.** Primary: **brain/CNS** (UBERON:0000955), **eye** (UBERON:0000970; retina UBERON:0000966; optic nerve UBERON:0000941), **heart** (UBERON:0000948; interventricular septum UBERON:0002094). Secondary/variable: **kidney/genitourinary** (UBERON:0002113), **ear** (inner ear UBERON:0001846), **palate** (UBERON:0001716).
 - **Body systems:** nervous, visual, cardiovascular, renal/genitourinary, auditory, craniofacial/skeletal.
 - **Tissue/cell level.** Nervous tissue (neurons; retinal ganglion cells CL:0000740), cardiac muscle, and **cranial neural crest**–derived mesenchyme (CL:0000333).
 - **Subcellular level.** Nucleus (GO:0005634); some variants show abnormal nuclear localization ([PMID: 36576487](#)).
 - **Localization / lateralization.** Anomalies are generally bilateral/midline (septal, palatal, optic fissure) reflecting disrupted developmental symmetry; RERE's WHHERE role in bilateral symmetry is notable ([PMID: 28959017](#)).
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8. Temporal Development

- **Onset:** congenital / early infancy. Structural anomalies are present at birth; DD/ID/ASD manifest in infancy–early childhood. Prenatal detection of associated malformations is possible (as in 1p36 deletion, where ventriculomegaly, cardiac defects, and dysmorphism prompt evaluation) ([PMID: 31172545](#)).
 - **Course:** the neurodevelopmental phenotype is generally **stable / non-progressive**, paralleling the "non-progressive neurocognitive syndrome" of the related ATN1/CHEDDA HX-motif disorder ([PMID: 30827498](#)). Rare progressive features (spasticity with Chiari I) are exceptional ([PMID: 41669386](#)).
 - **Duration:** chronic, lifelong.
 - **Critical periods:** embryonic organogenesis (retinal/optic-fissure closure, cardiac septation, palatal-shelf elevation, neurulation)—the windows during which RA/Shh dysregulation produces structural anomalies. This makes prenatal/perinatal windows the theoretical target for any future disease-modifying intervention.
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9. Inheritance and Population

- **Inheritance:** autosomal dominant. Variants are usually **de novo**; inherited transmission from a mildly-affected parent occurs, particularly for loss-of-function alleles ([PMID: 27087320](#); [PMID: 36053530](#); [PMID: 41988794](#)).
 - **Penetrance / expressivity:** highly **variable expressivity**; incomplete penetrance is suggested by mildly-symptomatic/asymptomatic parents carrying LoF variants.
 - **Epidemiology: ultra-rare.** Fewer than ~60 individuals are reported worldwide (10 in 2016; 23 by the 2022 review; 54 in the 2026 cohort including 30 previously unreported: "We assembled a cohort of 54 individuals with heterozygous pathogenic, likely pathogenic, and variants of uncertain significance in *RERE*, including 30 previously unreported cases"). No formal prevalence/incidence has been established. As an anchor, "Deletions of chromosome 1p36 affect approximately 1 in 5,000 newborns and are associated with developmental delay, intellectual disability, and defects involving the brain, eye, ear, heart, and kidney" ([PMID: 27087320](#); [PMID: 41988794](#)).
 - **Sex ratio / demographics:** no clear sex skew; cases span both sexes and diverse populations. No founder effect or consanguinity requirement (dominant, usually de novo). EHR studies suggest related rare conditions like 1p36 deletion may be underdiagnosed ([PMID: 39148290](#)).
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10. Diagnostics

Diagnosis is molecular. *RERE* variants are detected by **whole-exome sequencing (WES)**, **whole-genome sequencing (WGS)**, multigene neurodevelopmental/congenital-anomaly panels, or **chromosomal microarray (CMA)** (for 1p36 deletions encompassing *RERE*; FISH/karyotype for larger rearrangements) ([PMID: 27087320](#); [PMID: 41988794](#)).

No biomarker / no newborn screen. There is no biochemical or metabolic biomarker and no newborn screening test.

Supporting clinical work-up (phenotype-driven): brain MRI (structural CNS anomalies, Chiari I), ophthalmologic exam (coloboma, optic atrophy, refractive error), echocardiography (VSD/other CHD), renal ultrasound, audiologic assessment, and EEG for seizures.

Differential diagnosis and CHARGE overlap. In a WES study of 28 CHARGE-suggestive families, 53.6% had *CHD7* variants while "4 (14.3%) individuals had pathogenic variants in other genes (*RERE*, *KMT2D*, *EP300*, or *PUF60*)", and the authors concluded these "implicate a shared molecular pathology that disrupts epigenetic regulation of multiple-organ development" (PMID: 29300383). Accordingly, "Consideration should also be given to screening for *RERE* variants in individuals who fulfill diagnostic criteria for CHARGE syndrome but do not carry pathogenic variants in *CHD7*" (PMID: 29330883). Other differentials: 1p36 deletion syndrome, CHEDDA (*ATN1*), and other chromatin-regulator neurodevelopmental syndromes.

11. Outcome / Prognosis

- **Survival/mortality:** No formal survival statistics. Life-limiting risk derives mainly from **severe congenital heart defects** when present; the neurodevelopmental core is not itself lethal.
 - **Morbidity/disability:** Lifelong intellectual disability/developmental delay and ASD-related functional impairment dominate morbidity, with additional burden from epilepsy, vision/hearing deficits, and the surgical needs of structural anomalies.
 - **Disease course:** generally **stable/non-progressive** neurodevelopment; chronic lifelong condition.
 - **Prognostic factors: variant class is the key prognostic axis**—HRR missense/in-frame duplications predict more severe, multisystem disease; LoF variants predict milder outcomes (PMID: 29330883; PMID: 41988794). Presence and severity of CHD is the main determinant of early-life risk.
 - **QoL measures:** not formally applied in published cohorts.
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12. Treatment

No disease-specific or curative therapy exists. Management is supportive, multidisciplinary, and increasingly **variant-specific**: the 2026 cohort's findings "expand the clinical spectrum of *RERE*-related disorders, refine genotype-phenotype correlations, and support variant-specific" management (PMID: 41988794).

Domain	Intervention	Suggested NCIT
Developmental	Early-intervention, physical/occupational/speech therapy	Rehabilitation Therapy (NCIT:C15917)
Epilepsy	Anti-seizure medications	Anticonvulsant Agent (NCIT:C264)
Cardiac	Surgical repair of VSD/CHD	Cardiac Surgery (NCIT:C51648)
Orofacial	Cleft palate/lip repair	Cleft Palate Repair procedure
Ophthalmologic	Refractive correction, coloboma/optic-atrophy management	Ophthalmologic procedures
Audiologic	Hearing amplification/management of SNHL	Hearing Aid (NCIT:C50071)
Genitourinary	Management of renal/GU anomalies	Urologic procedures
Behavioral/psychiatric	ASD/ADHD supports; symptom-directed pharmacotherapy	Behavioral therapy; stimulants

Precedent from 1p36 comorbidity management: off-label melatonin (insomnia) and methylphenidate (ADHD) were effective without adverse effects in a 1p36 deletion case, illustrating symptom-directed pharmacotherapy relevant to overlapping neurobehavioral features ([PMID: 34831818](#)).

Experimental / preclinical lead. In zebrafish, the Shh inhibitor **HPI-1 rescues coloboma**, nominating Shh-pathway modulation as a candidate therapeutic direction ([PMID: 36576487](#)). No human clinical trials of targeted therapy exist. No pharmacogenomic guidance is established.

13. Prevention

- **Primary prevention:** none possible (usually de novo genetic disorder).
- **Secondary prevention / early detection:** phenotype-driven surveillance stratified by variant class—regular ophthalmologic, cardiac, audiologic, renal, and neurodevelopmental evaluation; brain imaging as indicated.

- **Genetic screening / counseling: genetic counseling** is central. Recurrence risk is low for confirmed de novo variants (with residual risk from germline mosaicism) but up to 50% for an affected transmitting parent; prenatal/preimplantation testing is feasible once a familial variant is known. Cascade testing of parents clarifies inheritance and recurrence risk (PMID: 41988794; PMID: 36053530).
- **Prenatal considerations:** as with 1p36 deletion, ultrasound findings (ventriculomegaly, cardiac defect, dysmorphism) may prompt CMA/sequencing (PMID: 31172545).
- **Immunization / public health / prophylaxis:** not applicable.

14. Other Species / Natural Disease

- **Taxonomy / orthologs:** *RERE* is highly conserved. Mouse *Rere/Atrophin-2* (*Mus musculus*, NCBI Taxon 10090); zebrafish paralogs *rerea* and *rereb* (*Danio rerio*, NCBI Taxon 7955). The zebrafish *rerea* mutant is named **babyface**.
- **Natural disease in other species:** No spontaneous companion-animal or wildlife NEDBEH-equivalent is catalogued (e.g., no established OMIA entry). Disease knowledge derives from engineered models rather than naturally occurring animal disease.
- **Comparative biology:** RA/Shh developmental roles and the RERE HX-repeat motif are evolutionarily conserved; the convergent human *ATN1* HX-motif disorder (CHEDDA) illustrates conserved motif-level pathogenicity (PMID: 30827498).
- **Transmission / zoonosis:** not applicable.

15. Model Organisms

Zebrafish (*Danio rerio*). The *rerea* mutant (**babyface**) "*robustly recapitulates optic fissure closure defects resulting from loss of RERE function, as observed in humans*"; coloboma results from expansion of the proximal optic stalk and reduced ventral retinal fate, and is rescued by the Shh inhibitor HPI-1 (PMID: 36576487). This model is powerful for eye phenotypes and for drug-rescue screening.

Mouse (*Mus musculus*). RERE-deficient and conditional mice recapitulate multiple organ-specific features: - Retinal/optic-nerve atrophy via retinal ganglion cell apoptosis from E17.5 (PMID: 33742727); - Ventricular septal defects via reduced GATA4 (PMID: 30061196); - Cleft

palate via cranial-neural-crest-specific ablation (Wnt1-Cre) (PMID: 33772547). - *Wdr5* and *Hdac1* mutant mice phenocopy RA-deficient asymmetric somite formation, validating the WHHERE complex mechanism (PMID: 28959017).

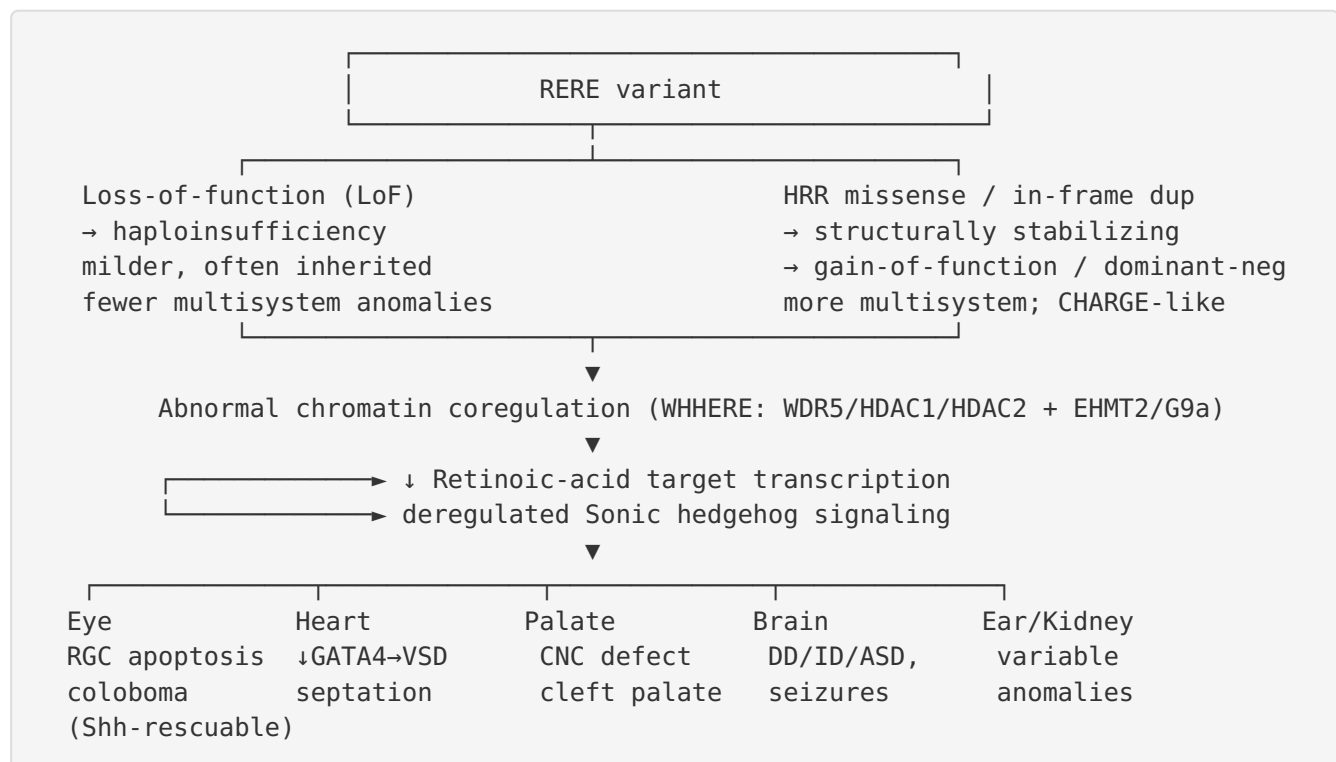
Genetic model types available: knockout, hypomorphic, and conditional (Cre-lox, e.g., Wnt1-Cre) mouse lines; zebrafish mutants.

Recapitulation & limitations. Models faithfully reproduce eye, heart, and palate anomalies and confirm the RA/Shh mechanism. Limitations: the human neurodevelopmental/behavioral phenotype (ID/ASD) is harder to model faithfully; models often assess loss-of-function, whereas HRR missense gain-of-function/dominant-negative biology is less fully modeled in vivo.

Resources: MGI (mouse), ZFIN (zebrafish).

Mechanistic Model / Interpretation

NEDBEH is best understood as a **transcriptional-coregulator developmental disorder** with a two-mechanism genetic architecture converging on RA and Shh dysregulation:



The variant-class axis is clinically actionable: it stratifies surveillance intensity and recurrence-risk counseling. The shared HX/HRR-motif biology across *RERE* and *ATN1* points to a broader class of "HX-motif perturbation" Mendelian syndromes. The Shh-inhibitor rescue in zebrafish is the single most promising translational lead.

Evidence Base

PMID	Study	Contribution
27087320	Fregeau 2016 (n=10)	Defined the syndrome; de novo RERE variants; RA coregulator; 1p36 overlap
29330883	Jordan 2018 (n=9)	Core genotype–phenotype correlation; CHARGE-like HRR duplication; RERE testing in CHD7-negative CHARGE
41988794	Curtis 2026 (n=54)	Largest cohort; LoF vs HRR biology; HRR missense stabilizing (GoF/ dominant-neg); variant-specific management
28959017	Vilhais-Neto 2017	WHHERE complex; RA-dependent symmetry; Pol II recruitment
36576487	Zebrafish babyface	Coloboma via shh deregulation; HPI-1 rescue (druggable lead)
33742727	Mouse eye	RGC apoptosis → retinal/optic atrophy
30061196	Mouse heart	↓ GATA4 → VSD
33772547	Mouse palate	Cranial-neural-crest cleft palate mechanism
29300383	Moccia 2018	RERE among CHARGE-overlapping chromatin genes; shared epigenetic pathology
30827498	ATN1/CHEDDA	Convergent HX-motif disorder; non-progressive
36053530	Case series	First inherited RERE variant; genotype–phenotype
41669386	Case report	Chiari I; spectrum expansion
25172301	1p36 microarray	Modifier genes; RERE and cardiac complications
31172545	Prenatal 1p36	Prenatal detection context
39148290	EHR study	Underdiagnosis of 1p36-related conditions
34831818	1p36 psychiatric	Symptom-directed pharmacotherapy precedent

Limitations and Knowledge Gaps

1. **Ultra-rarity (<60 cases)** limits precise phenotype frequencies, penetrance estimates, and formal prevalence/incidence.
 2. **No patient-level omics** (transcriptomics/proteomics/metabolomics) datasets; mechanism rests on model organisms and cell assays.
 3. **GoF/dominant-negative mechanism** of HRR missense variants is inferred from structural stabilization; direct in vivo functional proof is incomplete.
 4. **Neurodevelopmental phenotype modeling** is weak—models capture structural anomalies better than ID/ASD.
 5. **No QoL / natural-history longitudinal data**; survival and life-expectancy figures are unquantified.
 6. **Therapeutics** remain entirely supportive; the Shh-inhibitor lead is preclinical (zebrafish, eye only).
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Proposed Follow-up Experiments / Actions

1. **International registry & natural-history study** to quantify per-phenotype frequencies, penetrance, and progression by variant class.
2. **Functional validation of HRR missense variants** (knock-in mouse/zebrafish; degron/rescue assays) to confirm gain-of-function/dominant-negative mechanism vs LoF.
3. **Patient-derived iPSC neurons/organoids and cardiac/retinal organoids** for RERE variants to model neurodevelopmental and structural phenotypes and screen RA/Shh modulators.
4. **Preclinical Shh-modulation trials** extending HPI-1 rescue beyond the eye to cardiac/palatal endpoints, defining therapeutic windows.
5. **Chromatin/epigenomic profiling** (CUT&RUN/ChIP-seq for RERE, HDAC1/2, EHMT2; RNA-seq) in variant vs control cells to map dysregulated RA/Shh target genes.
6. **Variant-specific surveillance guideline** formalization and prospective evaluation, with structured genetic-counseling recurrence-risk pathways including germline-mosaicism considerations.

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